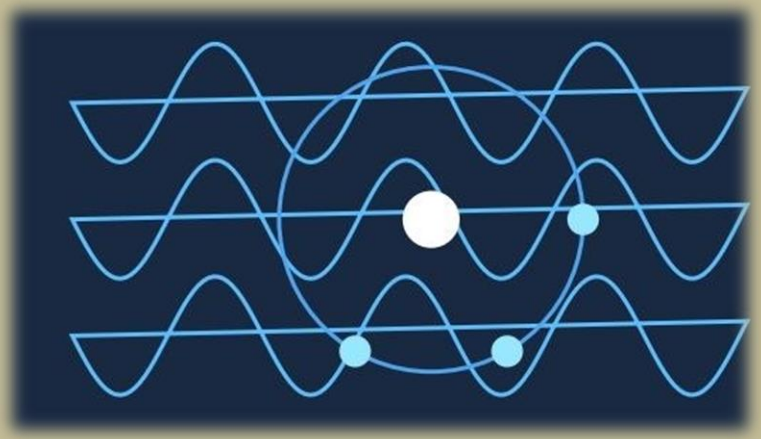
Nonlinear Optics

Lecture 1

Laser Engineering & Optoelectronics Department



Lect. Sujood Basim Jabr

Introduction to Nonlinear Optics

Nonlinear optics is a branch of optics that studies the interaction of intense light with matter, where the response of the material is not proportional to the applied electric field. These effects become important when high-intensity laser light interacts with a material. Nonlinear optical phenomena include second-harmonic generation, the optical Kerr effect, and four-wave mixing. Nonlinear optics has important applications in lasers, optical communications, spectroscopy, and modern photonic technologies.

Difference Between Linear and Nonlinear

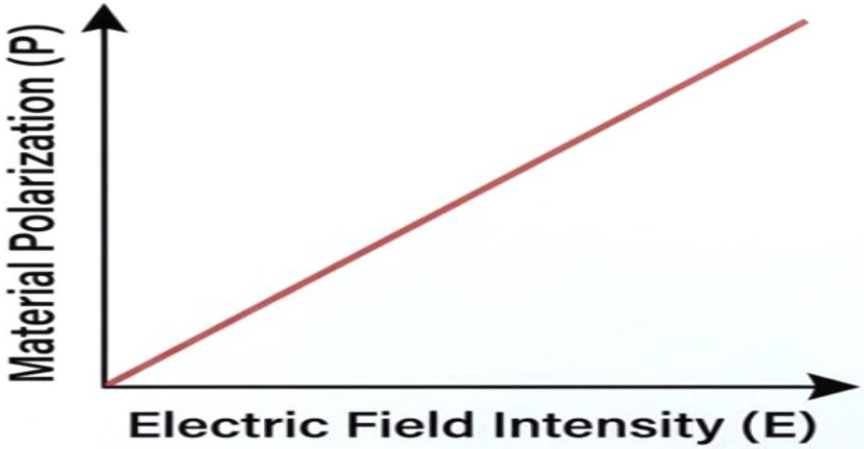
Linear Optics	Nonlinear Optics
The material response is proportional to the electric field.	The material response is not proportional to the electric field.
usually occurs at low light intensities	usually occurs at high light intensities, such as laser light.
The frequency of light generally remains unchanged.	New frequencies can be generated.
Examples: reflection, refraction, and interference.	Examples: second-harmonic generation, four-wave mixing, and the optical Kerr effect.

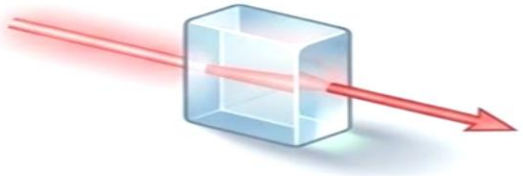
The response of the material to the electric field

- **The response of a material to an electric field** describes how the material reacts when an electric field is applied.
- **In linear optics:** the response is directly proportional to the electric field.
- **In nonlinear optics:** the response changes nonlinearly with the strength of the electric field.

Effect of Electric Field on Material

Linear Response

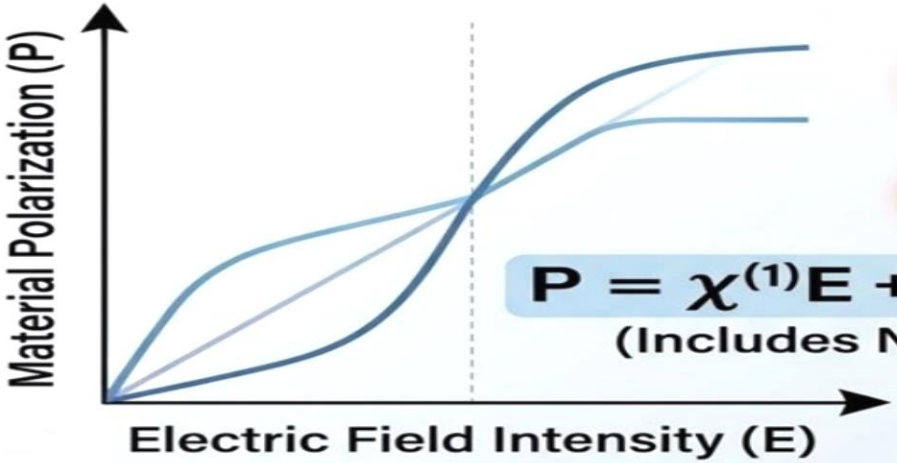


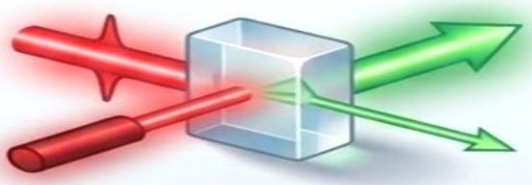


$$P \propto E$$

Directly Proportional

Nonlinear Response





$$P = \chi^{(1)}E + \chi^{(2)}E^2 + \chi^{(3)}E^3 + \ldots$$

(Includes Nonlinear Terms)

Activity Question:

What do you expect to happen to the optical response of a medium when the laser intensity becomes extremely high?



Time: 5 minutes

- **Linear polarization:**
  At low laser intensity, the polarization of a material is proportional to the applied electric field. The response is mainly linear and can be written as:
- $P=\varepsilon 0\chi(1)E$
- **Nonlinear polarization:**
  At very high laser intensity, the electric field becomes strong enough that the material response is no longer proportional to the field. Higher-order terms appear:
- $P=\varepsilon 0[\chi(1)E+\chi(2)E2+\chi(3)E3+\cdots]$

This nonlinear response can produce effects such as **second-harmonic generation, third-harmonic generation, and frequency mixing.**

Nonlinear Coefficient

The **nonlinear coefficient** describes how strongly a material responds to an electric field at high laser intensity.

It is commonly represented by:

$$\chi(2) \quad ,\chi(3)$$

$\chi^{(2)}$**:** Second-order nonlinear coefficient.

$\chi^{(3)}$**:** Third-order nonlinear coefficient.

A higher nonlinear coefficient means **stronger nonlinear optical effects** in the material.

Explain why:

A higher nonlinear coefficient means stronger nonlinear optical effects.



Problem: A laser beam has an optical intensity of:

I $= 1 \times 10^{10}$ W\ m^2 , Calculation of the Electric Field amplitude.

$$I = \frac{1}{2} c \varepsilon_0 E_0^2$$

حيث:

$$c = 3 \times 10^8 \ m/s$$

$$\varepsilon_0 = 8.85 \times 10^{-12} \ F/m$$

الحل:

نرتب المعادلة:

$$E_0 = \sqrt{\frac{2I}{c\varepsilon_0}}$$

بالتعويض:

$$E_0 = \sqrt{\frac{2(1 \times 10^{10})}{(3 \times 10^8)(8.85 \times 10^{-12})}}$$

$$E_0 \approx 2.75 \times 10^6 \ V/m$$

إذن:

$$\boxed{E_0 \approx 2.75 \times 10^6 \ V/m}$$


THANK YOU
FOR LISTENING
I APPRECIATE YOUR ATTENTION!